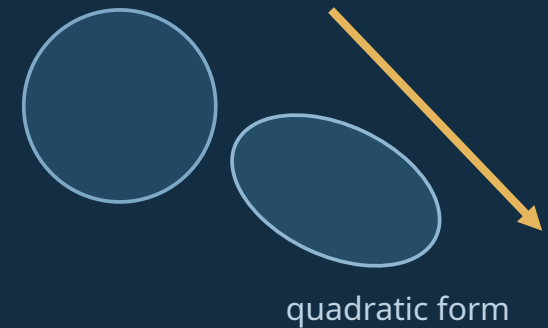


Multivariate Normal and Chi-squared Distributions

Distribution theory that powers least squares, sums of squares, and classical inference

$$Y \sim N_n(X\beta, \sigma^2 I) \implies \frac{SSE}{\sigma^2} \sim \chi_{n-p}^2$$



Learning objectives

The goal is not distribution theory in isolation: every result will be tied back to linear models.

ROADMAP

- Recognize and work with multivariate normal random vectors through means, covariance matrices, and linear transformations.
- Explain why zero covariance implies independence for jointly normal quantities, but not in general.
- Derive the chi-squared distribution as a sum of squared independent standard normal variables.
- Use projection matrices and quadratic forms to identify chi-squared distributions.
- Apply these results to $\hat{\beta}$, fitted values, residuals, SSE, nested-model sums of squares, and F tests.

The distributional chain in linear models

BIG PICTURE

1

Normal errors

ε / σ is a standard multivariate normal vector.

2

Linear transformations

$\hat{\beta}$, $\hat{Y}$, and e are linear transformations of Y .

3

Quadratic forms

SSE and model sums of squares are squared lengths of projections.

4

Chi-squared laws

Ranks of projection matrices become degrees of freedom.

5

Ratios

Independent chi-squared quantities lead to t and F inference.

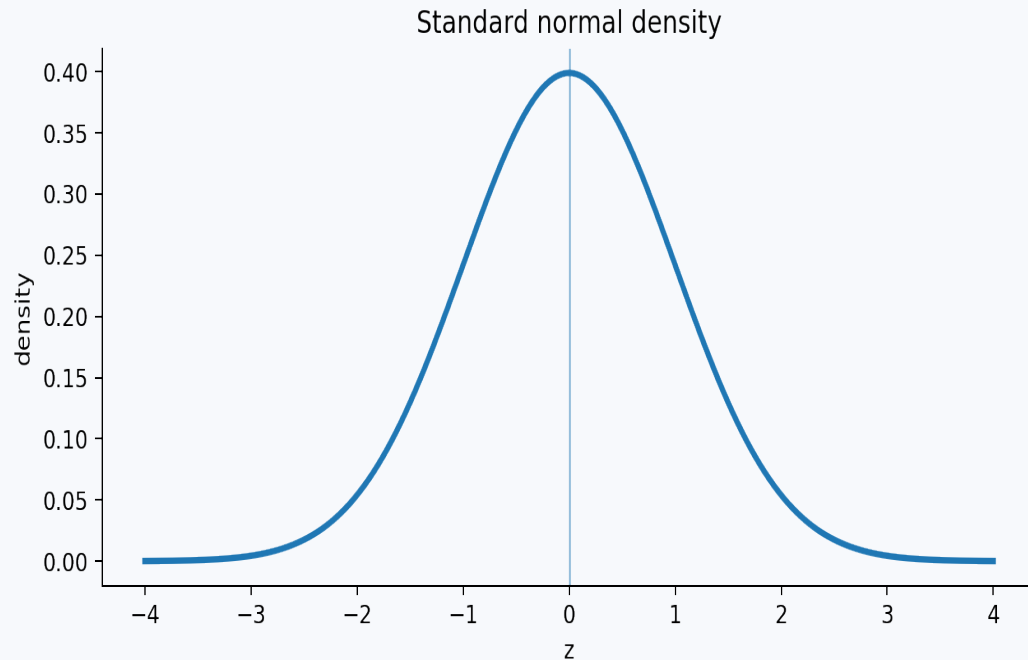
01

The multivariate normal distribution

From one normally distributed variable to an entire random vector.

Start with the univariate normal

NORMAL



Notation

$$Y \sim N(\mu, \sigma^2)$$

Standardization

$$Z = \frac{Y - \mu}{\sigma} \sim N(0, 1)$$

Linear-model preview

The multivariate version lets us standardize and transform an entire response vector at once.

A random vector

MVN

A random vector has:

Mean vector

Covariance matrix

$$\Sigma = \text{Var}(Y), \quad \Sigma_{ij} = \text{Cov}(Y_i, Y_j)$$

Dimensions

Y and μ are $n \times 1$; Σ is $n \times n$, symmetric, and positive semidefinite.

What does “multivariate normal” mean?

MVN

A random vector Y is multivariate normal if every linear combination is univariate normal:

$a^T Y$ is normal for every fixed $a \in \mathbb{R}^n$

We write $Y \sim N_n(\mu, \Sigma)$.

Why this definition is useful

Linear models are built from linear combinations. Once Y is multivariate normal, almost every estimator we construct inherits a normal distribution automatically.

The multivariate normal density

MVN

When Σ is positive definite, Y has density

$$f(y) = \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp \left\{ -\frac{1}{2} (y - \mu)^T \Sigma^{-1} (y - \mu) \right\}$$

Location

μ determines the center of the distribution.

Scale and orientation

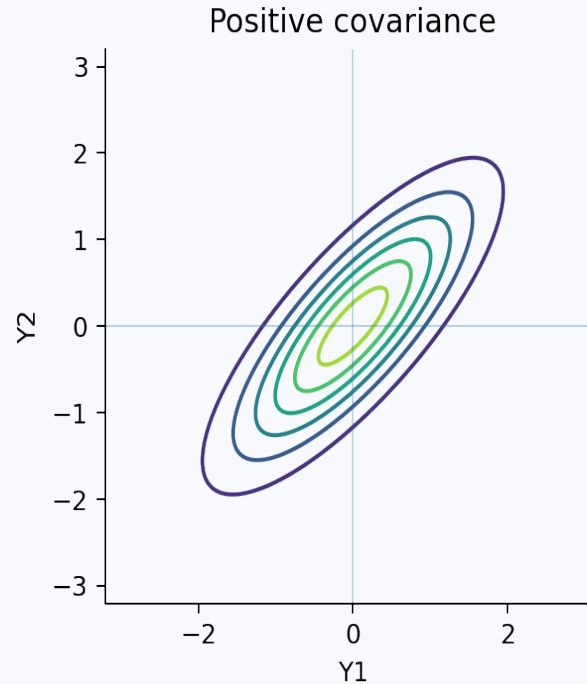
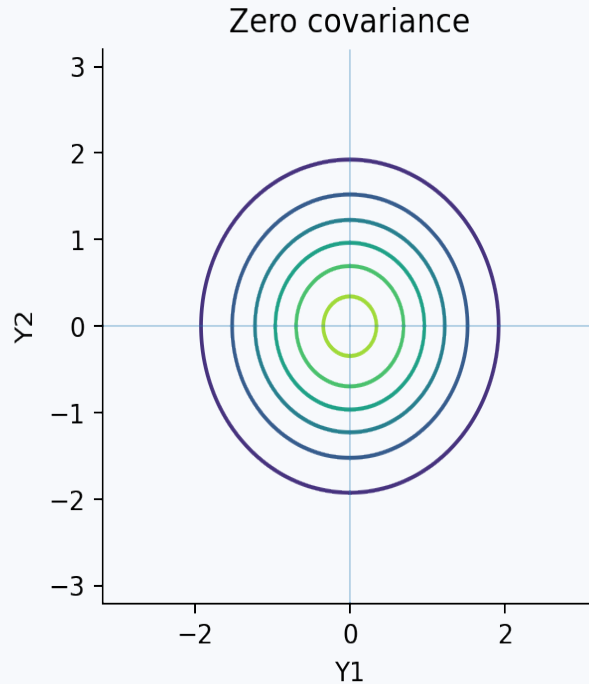
Σ determines marginal variances, correlations, and the geometry of the contours.

Distance

The exponent depends on Mahalanobis distance, not ordinary Euclidean distance.

Covariance controls the geometry

MVN



Zero covariance

For a bivariate normal distribution, zero covariance produces axis-aligned contours and independence.

Positive covariance

Positive covariance rotates the ellipses toward the line $y_1 = y_2$.

Marginal distributions are normal

MVN

If $Y \sim N_n(\mu, \Sigma)$,

then each component is normal:

$$Y_i \sim N(\mu_i, \Sigma_{ii}).$$

More generally, any subset of components is itself multivariate normal.

Linear-model connection

Any selected subset of responses, fitted values, residuals, or coefficients remains jointly normal whenever it is obtained by a linear transformation of Y .

The workhorse theorem: linear transformations

MVN

If $Y \sim N_n(\mu, \Sigma)$, and A is $m \times n$, then

$$AY + b \sim N_m(A\mu + b, A\Sigma A^T)$$

Means transform linearly

$$E(AY+b)=A\mu+b.$$

Covariances transform quadratically

$$\text{Var}(AY+b)=A\Sigma A^T.$$

Possibly singular

The transformed covariance need not be invertible; this occurs for fitted values and residuals.

Examples of linear combinations

MVN

Difference

Contrast

$$Y_1 - Y_2 = (1 \quad -1)Y$$

Average

Mean

$$\bar{\mathbf{Y}} = \frac{1}{n} \mathbf{1}^T \mathbf{Y}$$

Both are automatically normal when $\mathbf{Y}$ is multivariate normal.

The theorem replaces repeated distribution derivations with one matrix calculation.

Uncorrelated is not usually the same as independent

MVN

In general

$\text{Cov}(U,V)=0$ does not imply independence.

For jointly normal variables

$\text{Cov}(U,V)=0$ does imply independence.

This special property is essential in linear model theory.

$$\text{Cov}(AY, BY) = A\Sigma B^T = 0 \implies AY \perp BY \text{ (under joint normality)}$$

A block covariance view of independence

MVN

$$\begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} \sim N \left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \right)$$

If the off-diagonal block is zero

$\Sigma_{12} = 0$ implies the subvectors Y_1 and Y_2 are independent.

Linear-model preview

We will show $\hat{\beta}$ and e are jointly normal with zero cross-covariance. Their independence follows immediately.

The standard multivariate normal

MVN

$$Z \sim N_n(0, I_n)$$

means that $Z_1, \dots, Z_n$ are independent $N(0, 1)$ variables.

Why I matters

Identity covariance gives unit variance in every coordinate and zero covariance between coordinates.

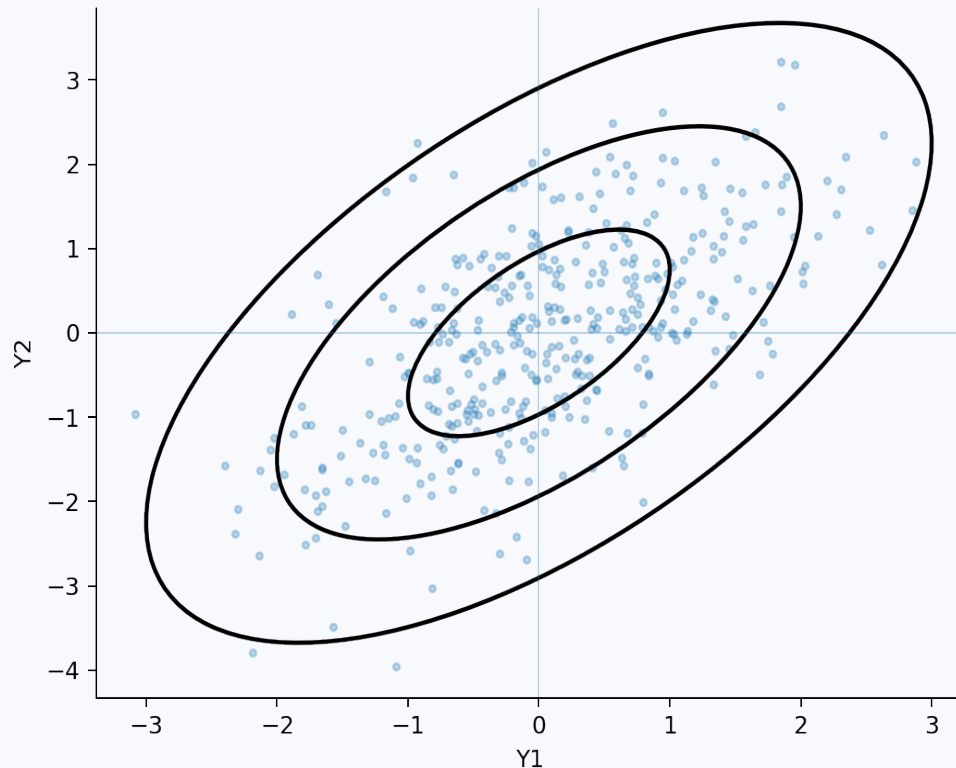
Where it comes from

In the normal linear model, $\varepsilon/\sigma \sim N_n(0, I)$, so all chi-squared results start from this vector.

Standardization in several dimensions

MVN $\rightarrow \chi^2$

Elliptical contours of constant Mahalanobis distance



For positive definite Σ , define the squared Mahalanobis distance

$$D^2 = (Y - \mu)^T \Sigma^{-1} (Y - \mu).$$

$$Y \sim N_n(\mu, \Sigma) \implies D^2 \sim \chi_n^2$$

This is the first direct bridge from the multivariate normal to the chi-squared distribution.

02

The chi-squared distribution

Squared lengths of normal vectors and the distribution of quadratic forms.

Start with one squared standard normal

CHI-SQUARED

$$Z \sim N(0, 1) \quad \Rightarrow \quad Z^2 \sim \chi_1^2$$

Why nonnegative?

Squaring removes the sign, so a chi-squared variable takes values only on $[0, \infty)$.

Why skewed?

Large absolute normal values become large positive squared values, producing a right tail.

One squared standard normal contributes one degree of freedom.

Definition of the chi-squared distribution

CHI-SQUARED

Let $Z_1, \dots, Z_v$ be independent standard normal variables. Then

$$Q = \sum_{j=1}^v Z_j^2 \sim \chi_v^2$$

Degrees of freedom

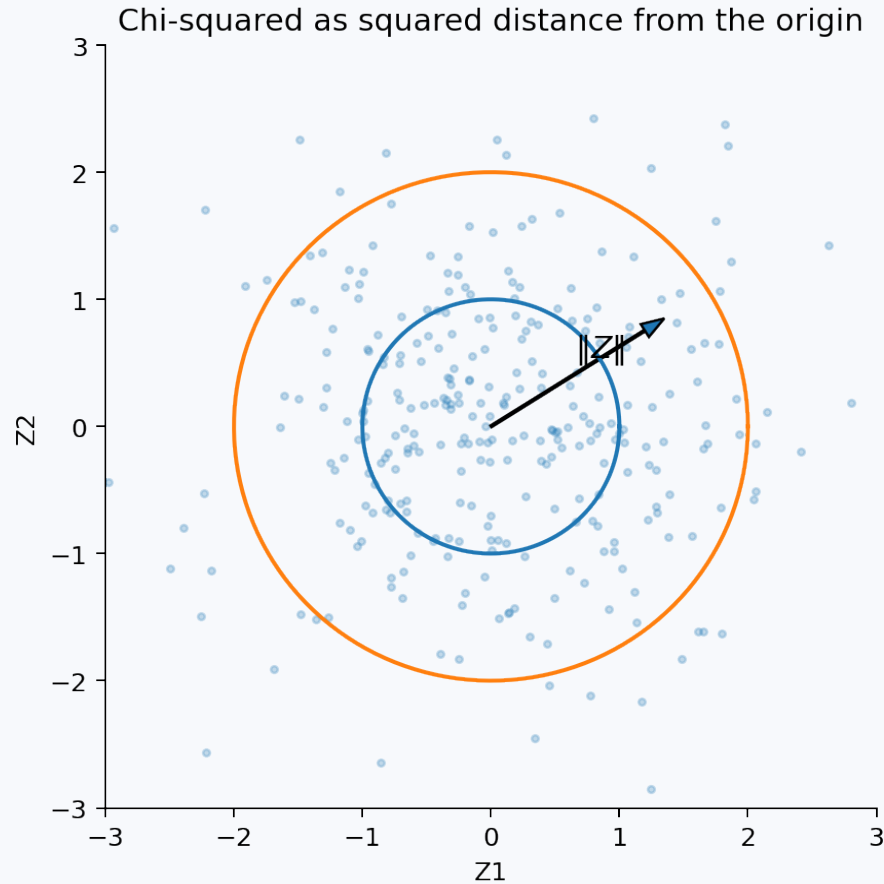
v counts the number of independent squared standard-normal directions contributing to Q .

Mean and variance

$E(Q)=v$ and $\text{Var}(Q)=2v$.

Geometric interpretation: squared distance

CHI-SQUARED



$$Z^T Z = \|Z\|^2 = Z_1^2 + \dots + Z_n^2 \sim \chi_n^2$$

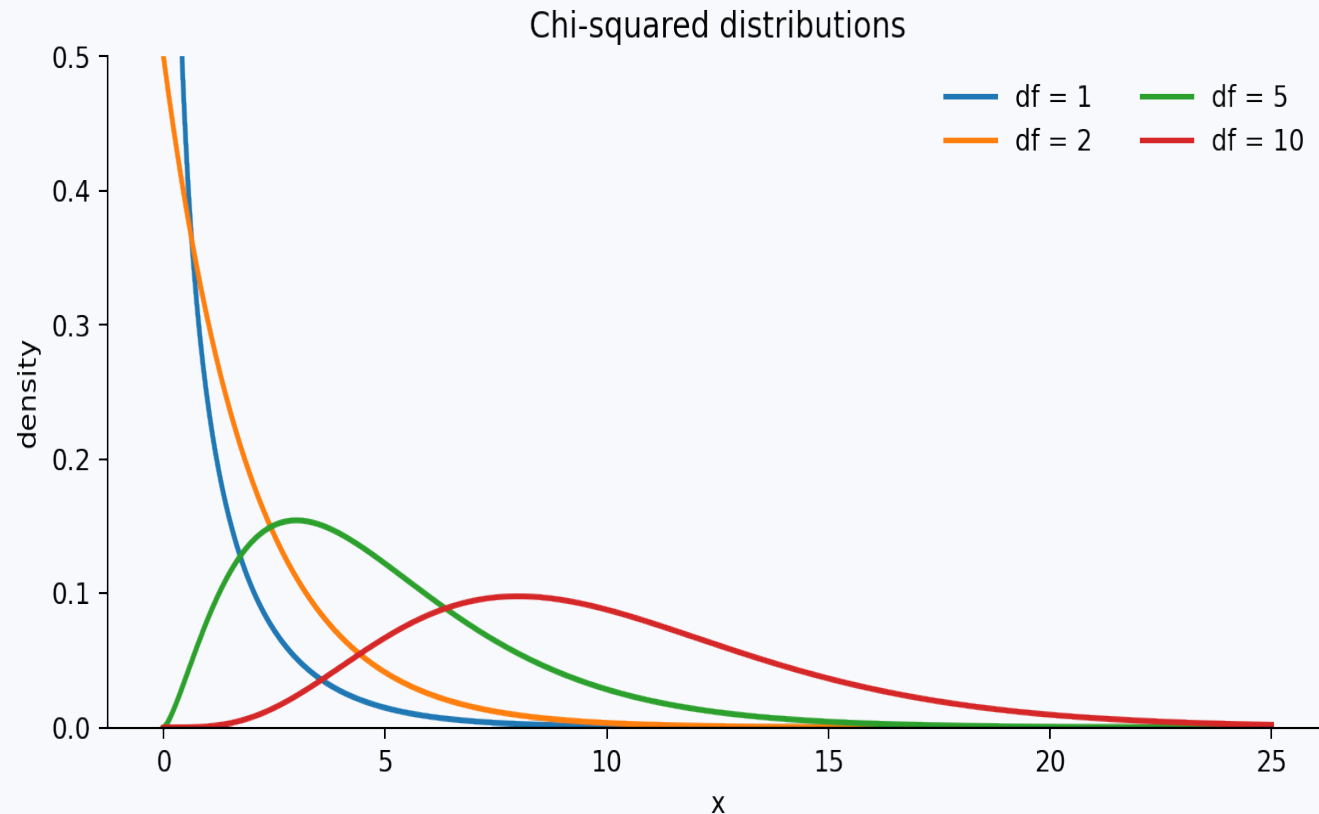
A chi-squared variable is the squared Euclidean length of a standard normal vector.

Degrees of freedom = dimensions

Each independent direction contributes one squared standard normal term.

How the shape changes with degrees of freedom

CHI-SQUARED



Small v

Strongly right-skewed; for $v=1$ the density is concentrated near zero.

Larger v

The distribution moves right and becomes more nearly symmetric.

In regression, v will usually be a matrix rank.

Additivity of independent chi-squared variables

CHI-SQUARED

$$Q_1 \sim \chi_r^2, \quad Q_2 \sim \chi_s^2, \quad Q_1 \perp Q_2$$

$$Q_1 + Q_2 \sim \chi_{r+s}^2$$

Geometric meaning

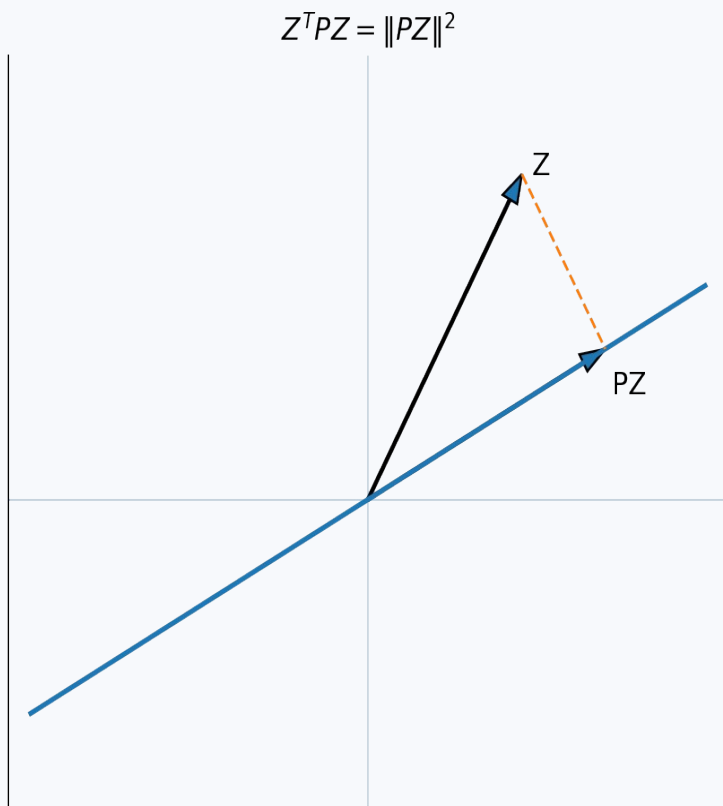
Independent orthogonal groups of normal coordinates contribute independent sums of squares, so their dimensions add.

Linear-model payoff

ANOVA decompositions are exactly decompositions into orthogonal sums of squares.

Projection matrices create chi-squared variables

QUADRATIC FORMS



Let $Z \sim N_n(0, I)$, $P = P^T = P^2$, $\text{rank}(P) = r$.

$$Z^T P Z = \|PZ\|^2 \sim \chi_r^2$$

Key interpretation

P keeps r orthogonal directions and annihilates the remaining $n-r$ directions. The quadratic form is the squared length of the retained normal component.

Why the projection theorem works

PROOF SKETCH

A symmetric idempotent matrix has eigenvalues only 0 and 1.

$$P = Q \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix} Q^T, \quad Q^T Q = I.$$

Rotate the standard normal vector: $U = Q^T Z \sim N_n(0, I).$

$$Z^T P Z = U_1^2 + \cdots + U_r^2 \sim \chi_r^2.$$

What rank does

Rank(P)=r counts the number of eigenvalues equal to 1 — exactly the number of squared normal coordinates that survive the projection.

More general normal quadratic forms

QUADRATIC FORMS

For symmetric A and $Z \sim N(0, I)$, spectral decomposition gives

$$A = Q\Lambda Q^T \implies Z^T A Z = \sum_{j=1}^n \lambda_j U_j^2.$$

Projection matrices are special

If A is idempotent, every λ_j is 0 or 1, so the weighted sum collapses to an ordinary chi-squared variable.

Otherwise

The quadratic form is generally a weighted sum of independent χ^2_1 variables, not a single chi-squared distribution.

What if the normal vector has a nonzero mean?

QUADRATIC FORMS

If $Y \sim N(\mu, \sigma^2 I)$ and P is a rank- r orthogonal projection, then

$$\frac{Y^T P Y}{\sigma^2} \sim \chi_r^2(\lambda), \quad \lambda = \frac{\mu^T P \mu}{\sigma^2}.$$

Central case

If $P\mu=0$, then $\lambda=0$ and the distribution is ordinary χ_r^2 .

Inference connection

Under a null hypothesis, the relevant projected mean is often zero; under the alternative it is not. This is the source of noncentral χ^2 and F power calculations.

03

The normal linear model

Now the pieces fit together: linear transformations give normals; projection quadratic forms give chi-squared laws.

The normal linear model

LINEAR MODEL

$$Y = X\beta + \varepsilon, \quad \varepsilon \sim N_n(0, \sigma^2 I)$$

$$Y \sim N_n(X\beta, \sigma^2 I)$$

Mean structure

The mean vector is restricted to the model space $\mathcal{C}(X)$.

Error structure

Independent errors with common variance correspond to covariance $\sigma^2 I$.

Full rank for now

Assume $\text{rank}(X)=p$ so $X^T X$ is invertible.

Distribution of the least-squares estimator

LINEAR MODEL

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

Because $\hat{\beta}$ is a linear transformation of a multivariate normal vector,

$$\hat{\beta} \sim N_p(\beta, \sigma^2 (X^T X)^{-1})$$

Unbiasedness

$$E(\hat{\beta}) = \beta.$$

Covariance

$$\text{Var}(\hat{\beta}) = \sigma^2 (X^T X)^{-1}.$$

Inference

Any linear contrast $c^T \hat{\beta}$ is univariate normal.

Fitted values and residuals are also normal

LINEAR MODEL

$$H = X(X^T X)^{-1} X^T, \quad M = I - H$$

Fitted values

$$\hat{Y} = HY \sim N_n(X\beta, \sigma^2 H)$$

Residuals

$$e = MY \sim N_n(0, \sigma^2 M)$$

Singular covariance is expected

H and M are projections, so their covariance matrices have ranks p and n-p rather than n.

Why $\hat{\beta}$ and the residuals are independent

LINEAR MODEL

They are jointly normal because both are linear transformations of Y .

Their cross-covariance is

$$\text{Cov}(\hat{\beta}, e) = \sigma^2 (X^T X)^{-1} X^T M = 0$$

$$X^T M = X^T (I - H) = 0$$

Joint normal + zero covariance

Therefore $\hat{\beta}$ and e are independent.

Why this matters

$\hat{\sigma}^2$ is a function of residuals, so $\hat{\beta}$ is independent of the residual variance estimator. This leads directly to t and F distributions.

The distribution of the residual sum of squares

SSE

$$SSE = e^T e = Y^T M Y = \varepsilon^T M \varepsilon$$

Standardize the error vector: $Z = \varepsilon / \sigma \sim N_n(0, I)$.

M is symmetric, idempotent, and $\text{rank}(M) = n - p$.

$$\frac{SSE}{\sigma^2} = Z^T M Z \sim \chi_{n-p}^2$$

This is the key chi-squared result

Residual degrees of freedom are literally the rank of the residual projection.

Why the residual degrees of freedom are $n - p$

DEGREES OF FREEDOM

$$\mathbb{R}^n = \mathcal{C}(X) \oplus \mathcal{C}(X)^\perp$$

Model space

$$\text{rank}(H) = p$$

Residual space

$$\text{rank}(M) = n - p$$

The p -dimensional fitted component uses p independent directions; the orthogonal residual component has $n-p$ directions left.

$$\text{tr}(H) = p, \quad \text{tr}(M) = n - p$$

Consequences for estimating σ^2

VARIANCE

$$\hat{\sigma}^2 = \frac{SSE}{n - p}$$

Since $SSE/\sigma^2 \sim \chi^2_{n-p}$,

$$E(SSE) = (n - p)\sigma^2$$

$$\boxed{E(\hat{\sigma}^2) = \sigma^2}$$

Exact finite-sample result

Normality gives the entire sampling distribution, not only unbiasedness.

$$\frac{(n - p)\hat{\sigma}^2}{\sigma^2} \sim \chi^2_{n-p}$$

Nested models: extra sums of squares

NESTED MODELS

Suppose a reduced model is nested inside a full model:

$$\mathcal{C}(X_R) \subset \mathcal{C}(X_F), \quad q = p_F - p_R$$

$$SSE_R - SSE_F = Y^T (H_F - H_R) Y.$$

Projection difference

$H_F - H_R$ is itself a symmetric idempotent matrix of rank q .

Under the reduced model

The mean lies in $\mathcal{C}(X_R)$, so $(H_F - H_R)X\beta = 0$.

$$\frac{SSE_R - SSE_F}{\sigma^2} \sim \chi_q^2$$

Orthogonal sums of squares are independent

NESTED MODELS

$$P_1 = H_F - H_R, \quad P_2 = I - H_F$$

These projections are orthogonal: $P_1 P_2 = 0$.

For a normal vector, orthogonal projected components are independent.

$$\frac{SSE_R - SSE_F}{\sigma^2} \perp \frac{SSE_F}{\sigma^2}$$

This is the engine behind the partial F test

One chi-squared measures the improvement from the added q -dimensional model space; the other estimates noise from the remaining residual space.

The F distribution emerges from two chi-squared variables

INFERENCE

If $U \sim \chi_q^2$, $V \sim \chi_\nu^2$, $U \perp V$, then $\frac{U/q}{V/\nu} \sim F_{q,\nu}$.

$$F = \frac{(SSE_R - SSE_F)/q}{SSE_F/(n - p_F)} \sim F_{q,n-p_F} \quad (H_0)$$

Numerator

Variation explained by moving from the reduced to the full model, per added degree of freedom.

Denominator

Residual variation under the full model, per residual degree of freedom.

General linear hypotheses fit the same pattern

INFERENCE

To test $H_0 : C\beta = d, \quad \text{rank}(C) = q,$

use the standardized normal contrast

$$C\hat{\beta} - d \sim N_q(0, \sigma^2 C(X^T X)^{-1} C^T) \quad (H_0)$$

Replace σ^2 by an independent estimate

Dividing by q and by $\hat{\sigma}^2$ converts the chi-squared statistic into the familiar general linear F test.

$$\frac{(C\hat{\beta} - d)^T [C(X^T X)^{-1} C^T]^{-1} (C\hat{\beta} - d)}{\sigma^2} \sim \chi_q^2.$$

Worked example: simple regression

EXAMPLE

Suppose $n=8$ and the model has an intercept and one slope, so $p=2$.

Residual projection

$$M = I - H, \quad \text{rank}(M) = 8 - 2 = 6$$

Residual sum of squares

$$\frac{SSE}{\sigma^2} = \frac{Y^T M Y}{\sigma^2} \sim \chi_6^2$$

If $SSE = 12$, then the unbiased variance estimate is

$$\hat{\sigma}^2 = \frac{12}{6} = 2.$$

Interpretation

Six orthogonal residual directions remain after estimating the two-dimensional mean structure.

Cochran's theorem: the unifying result

SYNTHESIS

Let $Z \sim N_n(0, I)$. Suppose symmetric idempotent matrices $P_1, \dots, P_k$ satisfy

$$P_i P_j = 0 \ (i \neq j), \quad P_1 + \dots + P_k = I.$$

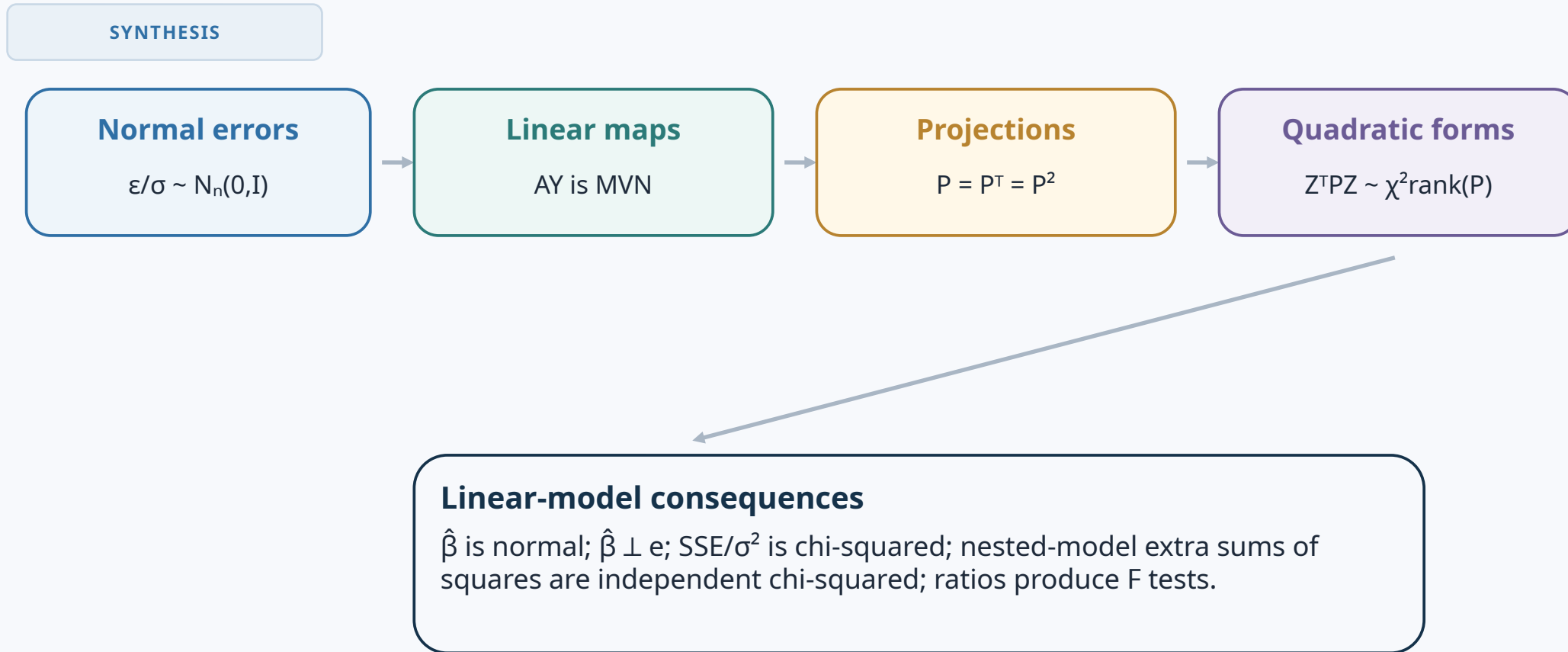
Then the quadratic forms $Z^T P_1 Z, \dots, Z^T P_k Z$

are independent, with $Z^T P_i Z \sim \chi^2_{\text{rank}(P_i)}$.

Linear models are a special case

H and M partition the observation space into fitted and residual subspaces. Nested-model projection differences partition it even further.

Concept map



What to remember

SUMMARY

- A multivariate normal distribution is stable under linear transformations.
- For jointly normal quantities, zero covariance implies independence.
- A chi-squared variable is a squared length of independent standard-normal coordinates.
- For a projection P , $Z^T P Z \sim \chi^2 \text{rank}(P)$; matrix rank becomes degrees of freedom.
- In the normal linear model, $\hat{\beta}$ is normal and independent of the residual vector.
- $SSE/\sigma^2 \sim \chi^2_{n-p}$, and orthogonal chi-squared sums of squares generate classical F tests.

In-class practice

PRACTICE

- 1 If $Y \sim N_4(\mu, \Sigma)$, what is the distribution of $a^T Y$ for a fixed $a \in \mathbb{R}^4$?
- 2 If $Z \sim N_6(0, I)$ and P is a rank-2 projection, what is the distribution of $Z^T P Z$?
- 3 In a normal linear model with $n=20$ and $p=5$, what is the distribution of SSE/σ^2 ?
- 4 Why are $\hat{\beta}$ and e independent? State the two ingredients needed for the argument.
- 5 A reduced model has $p_R=3$ and a full model has $p_F=6$. Under H_0 , how many degrees of freedom does the extra sum of squares have?